

Project Proposal

Vertical Jump Analysis and Classification via IMU Sensor Data

Group: Rudraksh, Parth, Himnish

Q. What is the project about?

This project aims to develop a wearable sensor system capable of accurately identifying and measuring vertical jumps while actively distinguishing them from other high-impact movements, such as heavy walking or running. By mounting an MPU6050 Inertial Measurement Unit (IMU) near the user's center of mass, we will capture 3-axis acceleration data. An ESP32 microcontroller will process this raw, noisy time-series data using low-pass filtering to calculate the total acceleration magnitude. We will implement algorithmic logic, starting with freefall (0g) time-thresholding and expanding into machine learning classification. This is to isolate the "flight phase" of a jump and calculate the exact jump height based on time-in-air.

Q. Why is the project exciting to you and your group?

This project bridges the gap between raw hardware signals and actionable data science. It is an exciting opportunity to build a real-world application similar to commercial sports science trackers. Instead of just writing software in a vacuum, we have to deal with the messy realities of the physical world, like sensor rotation, noise, and unexpected user movements. It allows us to apply our theoretical knowledge of algorithms, data visualization, and signal processing to a tangible physical device.

Q. What new hardware do you need besides what has already been provided?

1. Extra-long USB-to-MicroUSB cable to attach the device to the laptop for data recording
2. Velcro straps or a 3D-printed belt clip to firmly secure the ESP32 and MPU6050 to the user's lower back/waist to minimize independent sensor movement.

Q. Who are the team members and what are their roles?

Rudraksh → Data & Algorithm Developer:

Responsible for the software pipeline. Will handle the ingestion of raw MPU6050 sensor data via serial communication using C and Python. Will implement noise filters and develop the jump-detection logic. Will explore training classification models to separate walking features from jumping features, and utilize Matplotlib and Seaborn for data visualization and system validation.

Himnish → Test Engineer

Responsible for designing the experimental testing protocol. Will manage the tethered data collection sessions (recording datasets of varied walking, running, and jumping while managing the cable), manually label the datasets for algorithmic training, calibrate the physical threshold variables, and oversee the physical mounting of the hardware.

Parth → Embedded Systems and visualizations and report generation

Responsible for wiring the physical components and establishing reliable I2C communication between the MPU6050 and the ESP32. Will configure the ESP32 to accurately stream data over the serial port at a high sampling rate to the host laptop. Use Matplotlib and Seaborn for data visualization and system validation and to generate reports summarizing the findings.

Q. What are the milestones and deliverables?

Milestone 1 (Hardware Setup): Successfully wire the MPU6050 to the ESP32 via I2C and stream raw X, Y, Z acceleration data to the laptop.

Milestone 2 (Data Collection & Filtering): Record baseline datasets of jumping. Implement the magnitude calculation and low-pass filtering to clean the raw data in Python.

Milestone 3 (Algorithm Development): Develop the initial threshold-based algorithm to successfully detect a jump's free-fall phase and calculate height.

Milestone 4 (Refinement & Edge Cases): Implement classification logic to eliminate false positives (e.g., aggressive walking).

Milestone 5 (Final Deliverable): A live demonstration of the sensor accurately logging jump heights in real time as well as submission of the final report and code.

Q. What is the project's GitHub link?

https://github.com/himccoder/IMU_Jumplab.git